

A Comparison of Input Devices for Precise Interaction Tasks in VR-based Surgical Planning and Training

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ABSTRACT

We present a comparison of input devices for common interaction tasks in medical VR training and planning based on two relevant applications. The chosen devices, VR controllers, VR Ink, data gloves, and a real medical instrument, differ in their degree of specialization and their grip. The conducted user study shows that the controllers and VR Ink performed significantly better than the other devices regarding precision. Concerning questionnaire results, no device stands out but most participants preferred the VR Ink for both applications. These results can serve as a guide to identify an appropriate device for future medical VR applications.

Index Terms: Computing methodologies—Computer graphics—Graphics systems and interfaces—Virtual reality; Human-centered computing—Human computer interaction (HCI)—Interaction devices; Human-centered computing—Human computer interaction (HCI)—HCI design and evaluation methods—Usability testing;

1 MOTIVATION

Virtual reality (VR) is widely used for medical applications because it benefits from direct interactions with 3D models without harming patients or consuming resources. For these applications, the right choice of input device based on the application's tasks and purpose is crucial. Depending on the tasks, some devices might be more supportive and efficient due to the corresponding hand position. Furthermore, the purpose of the application also plays a role. On the one hand, a general device benefits from its flexibility and can be used for several different tasks. On the other hand, a highly specialized device that is similar to a real medical instrument, is crucial for medical training applications.

For medical VR applications, the ways of interacting and input devices range from hand gestures [6], standard VR controllers [1, 15] to stylus devices [2, 9, 12] to highly specialized medical devices for training applications [8, 11]. To assess benefits and drawbacks, several studies have been conducted. Some of them compared the precision of devices held in power grip and precision grip [4, 5, 14] for pointing or sketching tasks. Further studies compare gestures and conventional interaction [10] or gestures vs. VR controllers [13]. In our comparison, we use devices held in power grip and precision grip as well as hand gesture interactions and additional tasks that are relevant for medical applications such as rotation. Furthermore, we investigate the perceived usability besides precision and cover different medical applications and thus interaction tasks.

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In this work, we conducted a user study to compare different input devices for medical applications using two relevant and representative use cases: a liver surgery planning application and a craniotomy training application. A craniotomy is part of the majority of brain surgeries, whereas liver surgery planning comprises the creation of an incision line and resection plane which can easily be applied to other tumor resection surgeries. Additionally, the comprised interactions and tasks such as selecting, grabbing, scaling and rotating are often part of exploring anatomical structures. The included task of precisely drawing is similar to following a trajectory. This is important for medical tasks like navigating in air-filled structures or cutting which is part of every surgery.

2 STUDY DESIGN

For the study we used the two representative medical applications which are inspired by a liver surgery planning system and a craniotomy training system proposed by Chheang et al. [7] and Allgaier et al. [3] respectively.

Input Devices The tasks as well as the purpose of the applications affect the choice of the input device. Our comparison comprises four input devices in total, three per application. For the liver surgery planning we used the VR controllers, VR Ink and data gloves, whereas for the craniotomy training we used VR controllers, VR Ink and a craniotome, which is the real medical instrument.

The VR controllers serve as the state of the art. Head-mounted displays always involve controllers. Thus, they are easily available and cheap as well as familiar to people using VR frequently. Although there are small differences between VR controllers, they all have the power grip in common. The second device used in both applications is the VR Ink (Logitech, Switzerland). This device was chosen because one of the main tasks in both applications is to draw, and the natural hand position for precise drawing cannot be provided by a controller but by a pen-like device such as the VR Ink. Especially in the craniotomy training application, a correct and realistic hand position during performing the craniotomy is crucial to build up correct muscle memory. With the Manus data gloves (Manus VR, Netherlands), we included a device for intuitive hand interaction, especially for tasks like pointing and scaling. To avoid the mentioned incorrect muscle memory problem, we decided to use the gloves only for the liver surgery planning application and not for training purposes, as they differ the most from the real device. Consequently, for the craniotomy training we chose the best device regarding muscle memory, the craniotome itself.

Task Description Both applications comprise the interactions selecting and pointing when interacting with the user interface. Precisely drawing the incision line midair is equivalent to the planning using 3D printed models or highlighting the incision line on 2D image data. In the craniotomy training, precisely drawing the craniotomy contour simulates the procedure of cutting the bone. Here it is crucial to be able to draw a very precise contour to enable proper

access without harming the patient. Additionally, the liver model can be scaled and in the craniotomy training application, the patient's head can be rotated via hand rotation and the blood vessels can be rotated for exploration via trackpad. For scaling the liver, two devices are necessary, thus the VR Ink is used with a controller in the second hand.

3 EVALUATION

The evaluation was conducted with 22 participants, including four medical experts. The study was designed in a way that medical expertise is not necessary. During and after performing the application specific workflow with all three devices in a randomized order, the participants had to answer questionnaires concerning task load, cybersickness and usability. Additionally, quantitative data, such as task completion time, drawing resets, drawing and deforming attempts, deforming error, precision of drawing a given incision line and craniotomy contour, and Hausdorff distance were measured and calculated.

4 RESULTS AND DISCUSSION

To analyze the obtained data and to see whether there are significant differences between the input devices with regard to the user performance, one-way repeated measures ANOVA were conducted. Additionally, the post-hoc analyses using pairwise t-test with *Bonferroni* correction to compare between the input devices were performed after the results of significant effects. In the following, the significant results are presented.

For both applications we found statistically significant effects on drawing precision ($p = 0.0034$). Regarding the liver surgery planning application, there are significant differences between the controller and the data gloves ($t = 3.37$, $df = 21$, $p < 0.009$), and between the VR Ink and the data gloves ($t = -3.48$, $df = 21$, $p < 0.007$). For the precision in the craniotomy training application ($p < 0.00161$) post-hoc analysis revealed significant differences between the controller and the craniotome ($t = 2.73$, $df = 21$, $p < 0.038$), and between the VR Ink and the craniotome ($t = -3.72$, $df = 21$, $p < 0.004$).

We also found significant differences between the devices when analyzing the suitability for the tasks selecting and pointing as well as drawing. Regarding selecting and pointing, the post-hoc analysis shows statistically significant differences between the controllers and data gloves ($p < 0.0103$) and between the VR Ink and data gloves ($p < 0.0065$). There was no statistically significant difference between the controllers and VR Ink. The craniotome cannot be used for this task. For the drawing interaction, we found statistically significant differences between the VR Ink and controllers ($p = 0.011$), between the VR Ink and data gloves ($p < 8.3e^{-5}$), and between the VR Ink and craniotome ($p < 1.6e^{-6}$).

5 CONCLUSION

Concluding all results, the VR Ink and controllers do not show statistically significant differences, except for the drawing task, where the VR Ink performs significantly better. Nevertheless, for most participants the VR Ink is the preferred device. This overall ranking coincides with the results from the usability questionnaire, where the VR Ink always reached slightly better results than the controller. From this, one can conclude that they felt confident using this device.

The results reveal the benefits and drawbacks of the different devices for precise interaction tasks in VR-based surgical planning and training and can provide good assistance when selecting an appropriate device for specific medical tasks.

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